# The Determination of a Real/Complex Matrix Condition Number Based on it's Sub Matrices

Investigation of how the condition numbers change for matrices of random numbers as the matrix dimensions gets larger, leading to a remarkable formula (and several subsequent inequalities) that computes the exact condition number based on sub-matrices.

Proofs are included.

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# Summary of Results

The main result is the formula below that relates the (Frobenius) condition number of a matrix to the submatrices it contains. This knowledge can be used to trace a large condition number (ie: a matrix instability) to certain values and submatrices. Research shows that in most tests of random matrices with very large cond numbers the reason for this lies inside the structure of:

$$L_n = I_n + m_0 v_0 A_n^{-1}$$

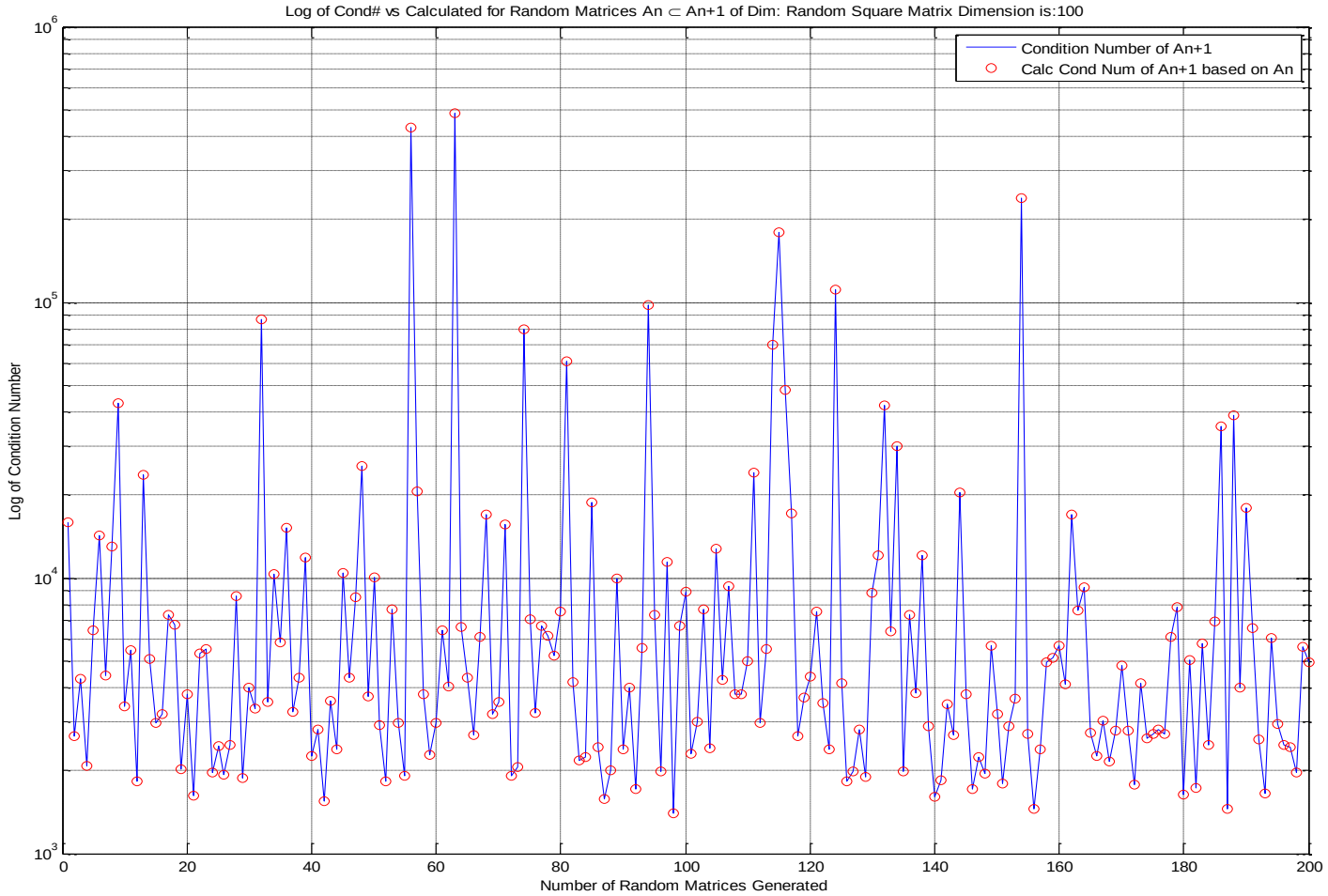
Theorem 6.2

Let: $b_0 = b_{n+1,n+1}$; $k_0 = V_2 A_n^{-1} V_1$; $m_0 = 1/(b_0 - k_0)$, $V_1 V_2 = V$, $L_n = I_n + m_0 V A_n^{-1}$

Then: $\text{cond}(A_{n+1}) = \| A_{n+1} \| \circ \| A_{n+1}^{-1} \| =$

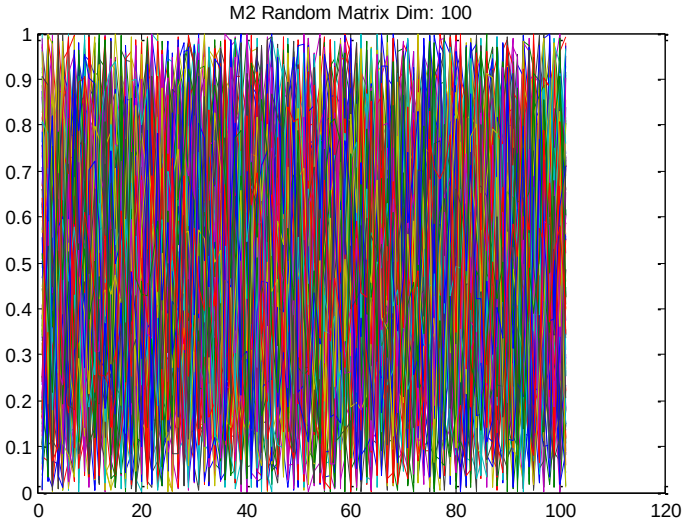
$$\sqrt{\| A_n \|^2 + \| V_1 \|^2 + \| V_2 \|^2 + | b_0 |^2} \sqrt{\| A_n^{-1} L_n \|^2 + | m_0 |^2 (\| A_n^{-1} V_1 \|^2 + \| V_2 A_n^{-1} \|^2 + 1)}$$

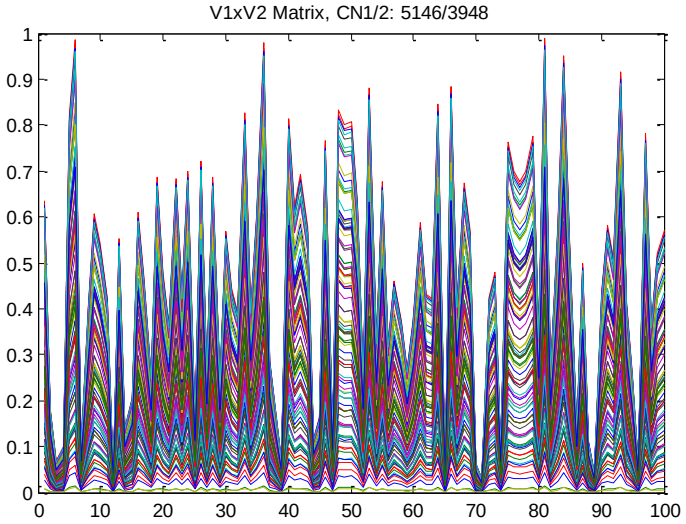
# Numerical ‘Proof’ of Results

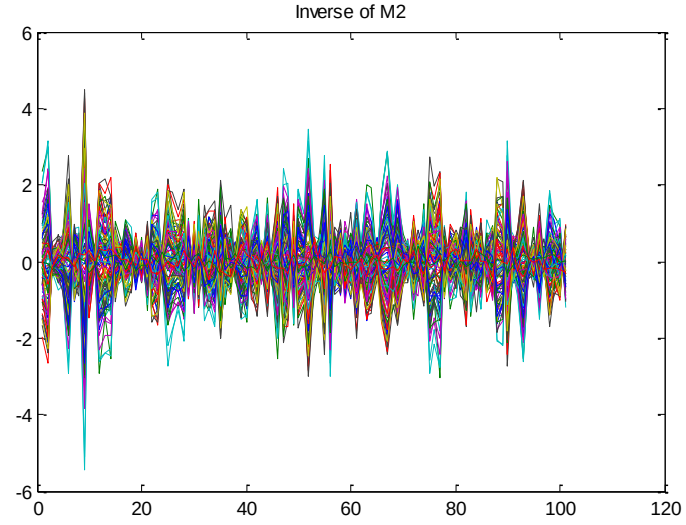


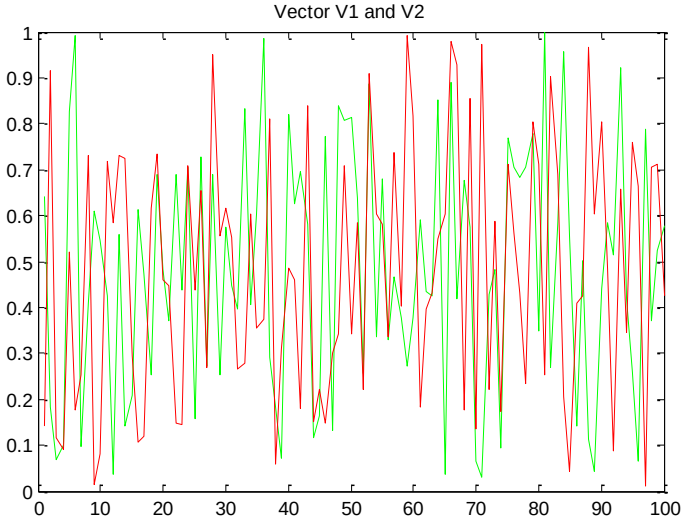

Above we see how the formula coincides with the computed cond(A,’fro’)

The circles coincide with all the peaks/valleys which are the cond numbers going up and down as the 200 random 100x100 matrices are generated



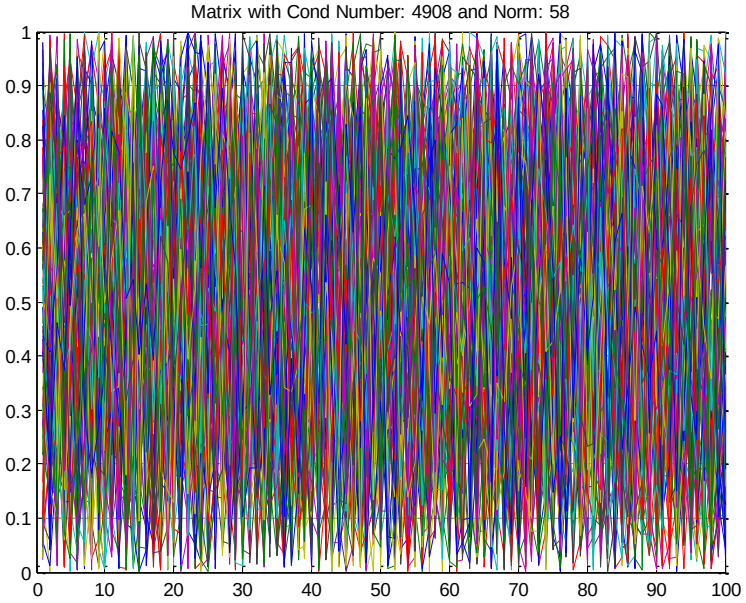

The matrix. inverse, submatrices and edge vectors with cond num of 5145 in a 100x100 random matrix

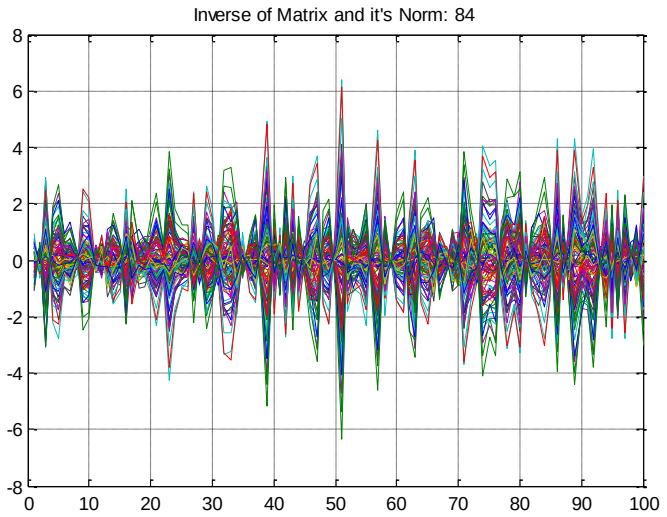
# An example of a $100^2$ matrix with 'large' condition number of 4,908 and it's decomposition process

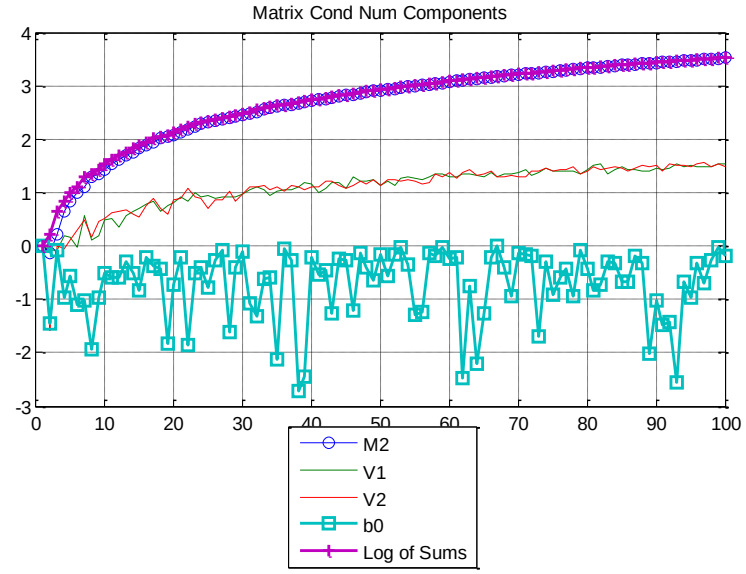
This shows the factors of the condition number per dimension of the submatrices
An, V1,V2 and m0.
The factors are the terms in:

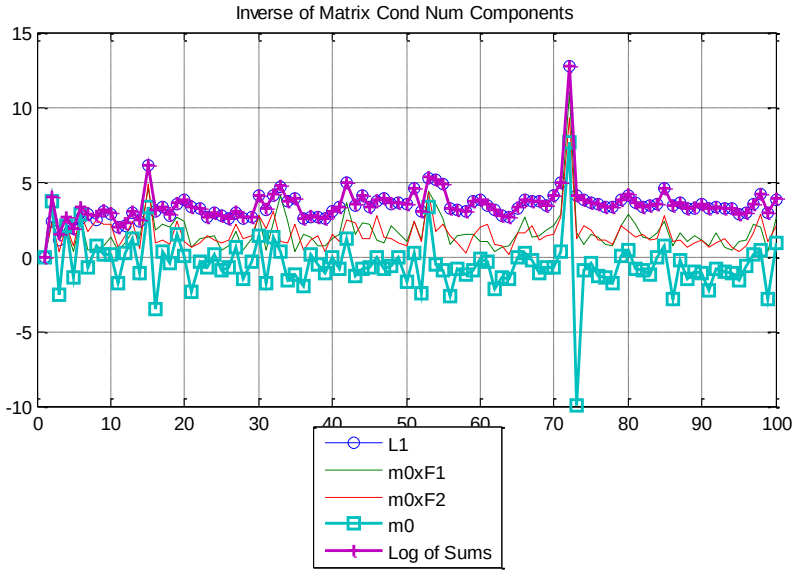
$$\mathrm{cond}(A_{n+1}) = \| A_{n+1} \| \circ \| A_{n+1}^{-1} \| =$$

$$\sqrt{\| A_n \|^2 + \| V_1 \|^2 + \| V_2 \|^2 + | b_0 |^2} \sqrt{\| A_n^{-1} L_n \|^2 + | m_0 |^2 (\| A_n^{-1} V_1 \|^2 + \| V_2 A_n^{-1} \|^2 + 1)}$$


Matrix with Cond Number: 4908 and Norm: 58


Inverse of Matrix and it's Norm: 84


Matrix Cond Num Components
M2
V1
V2
b0
Log of Sums


Inverse of Matrix Cond Num Components
L1
m0xF1
m0xF2
m0
Log of Sums

# Simple program to find matrices $100^2$ with large cond numbers > 100,000,000

```
%  RAUL GARCIA, 10-20-2010
%  FOR 100X100 RANDOM MATRICES

clear all

large=100000000;
%  ------------ SEEK COND NUM ABOVE THIS NUMBER
disp 'Seeking 100x100 matrices over this cond# ',num2str(large)

d=100;
trials=10000;
ix=1:trials;


disp 'Finds a dim = ',num2str(d)
disp 'square matrix with large cond number ',num2str(large)


for x=ix
    m=rand(d);
    if cond(m,'fro')>large
       disp 'Matrix saved as condMatrix'

       condMatrix=m;
       condMatrix
       condNumber=cond(condMatrix)
       disp '...'
       disp 'try#, matrix, cond number'
       x

    end
end
```

# Observations for Large Condition Numbers

- When generating thousands of random matrices of dimension $100^2$ to $500^2$ etc we selected those with 'large' condition numbers (see previous charts).
- The pattern we saw before was very common: the matrix condition number growth was log-stable, with no jumps for the 1st factor: $\sqrt{\| A_n \|^2 + \| V_1 \|^2 + \| V_2 \|^2 + | b_0 |^2}$
- Now the pattern for the inverse of the matrix had usually one large jump, often in the previous submatrix, we can see this on the previous charts on the matrix decomposition for the factors of:

$$\sqrt{\| A_n^{-1} L_n \|^2 + | m_0 |^2 \, (\| A_n^{-1} V_1 \|^2 + \| V_2 A_n^{-1} \|^2 + 1)}$$

# Related Result and Reference

- There seems to be few fundamental results on condition numbers
- The link below is to a classic Edelman (MIT 1989) thesis paper on condition numbers that tries to answer the **key question** (coming from Neumann, Smale, Birkhoff and others) is:

- **'What is the condition number of a real/complex Random Matrix?'**

- The author believes that the current results truly answer this question.
- Edelman paper relies on statistical results on the growth of random matrices.
- **The current paper gives a 100% *deterministic* result on the nature of the Frobenius condition number – the most used in numerical linear algebra**
- We also show how the same result can recursively be applied to the submatrices, resulting in a result for condition numbers similar to the prime factorization theorem for integers, since we 'factor out' a condition number completely.
- **In slide 61 we explain that a result for ALL matrices (like this one) can be applied for any ill-conditioned matrix in engineering and any other applied mathematics field. This is explored in the last part of this research.**
- **We use random matrices just to produce matrices and test numerically our results and for finding examples of matrices with large condition numbers.**

- Link and pdf on Edelman paper
- http://www-math.mit.edu/~edelman/homepage/papers/Eig.pdf

# Open Questions
## Some Possible Directions of Research

$$L_n = I_n + m_0 V A_n^{-1}$$

The matrix above seems the cause of much of the matrices ill-conditioning or condition number explosive growth.

Since the identity matrix is involved there seems to be here some relation to eigenvalues – since $L_n$ is a sum of *I* and a multiple of the inverse of An. Then this matrix is multiplied in the left-side by the inverse again in the complete formula.

If we see the origin of $L_n$ at right we see exactly how the last diagonal, the edge vectors and the inverse matrix define the norm of the inverse of A, a factor of the condition number $||A||\ ||A^{-1}||$.

Let: $b_0 = b_{n+1,n+1}$

$$k_0 = V_2 A_n^{-1} V_1$$

$$m_0 = 1/(b_0 - k_0)$$

$$V_1 V_2 = V$$

$$L_n = I_n + m_0 V A_n^{-1}$$

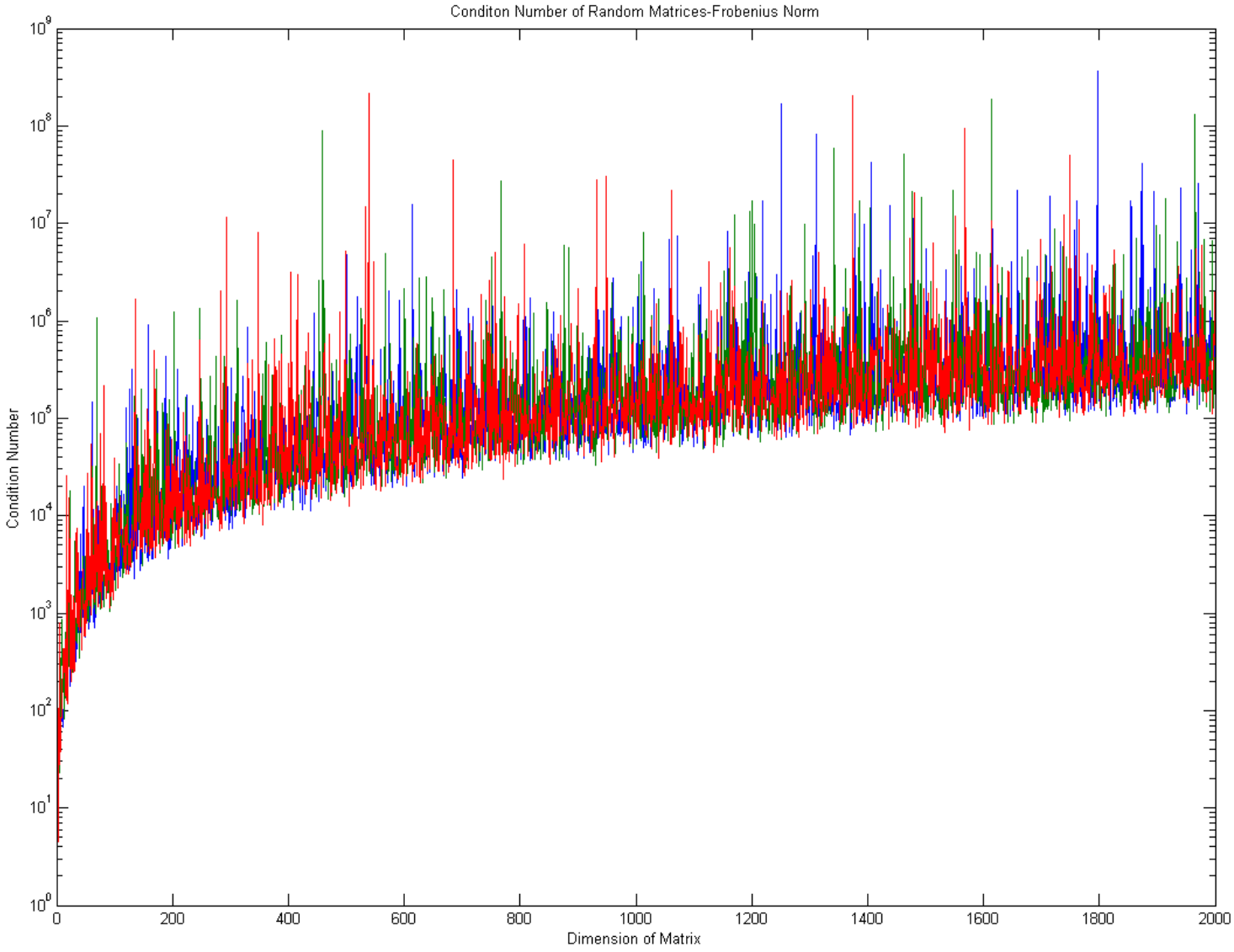
In the next slides we see how as a random matrix increases in dimension the condition numbers also increase beyond a certain lower bound.

This is explored in Edelman's paper.

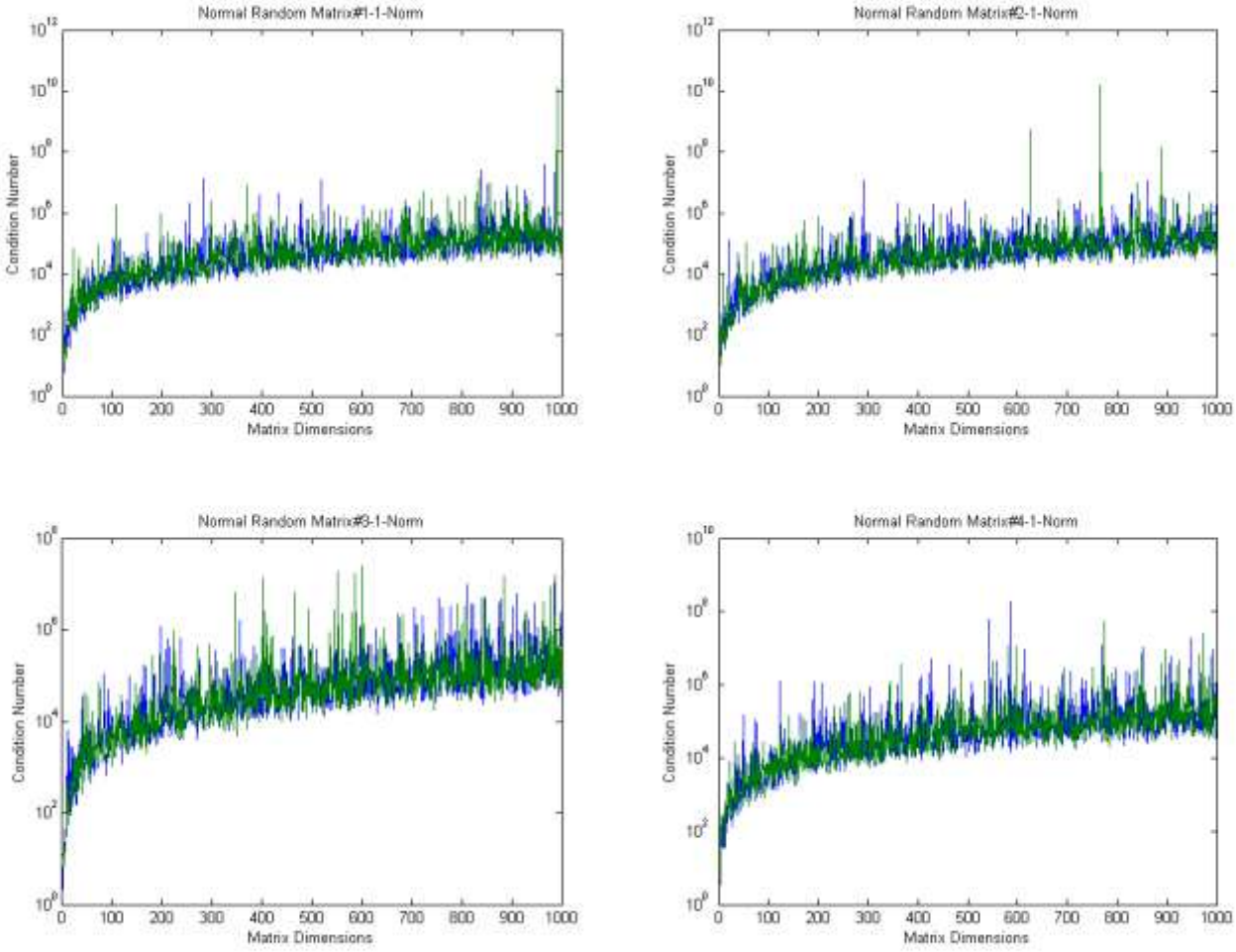
Next we see how this is independent of the norm used and the type of random distribution used.
We try with uniform vs normal distributions.

All the charts are log in y, the x-axis is the square matrix dimension and y the log of the condition numbers.

Single color plots are for a single random matrix per dimension, the color are more than 1 matrix.

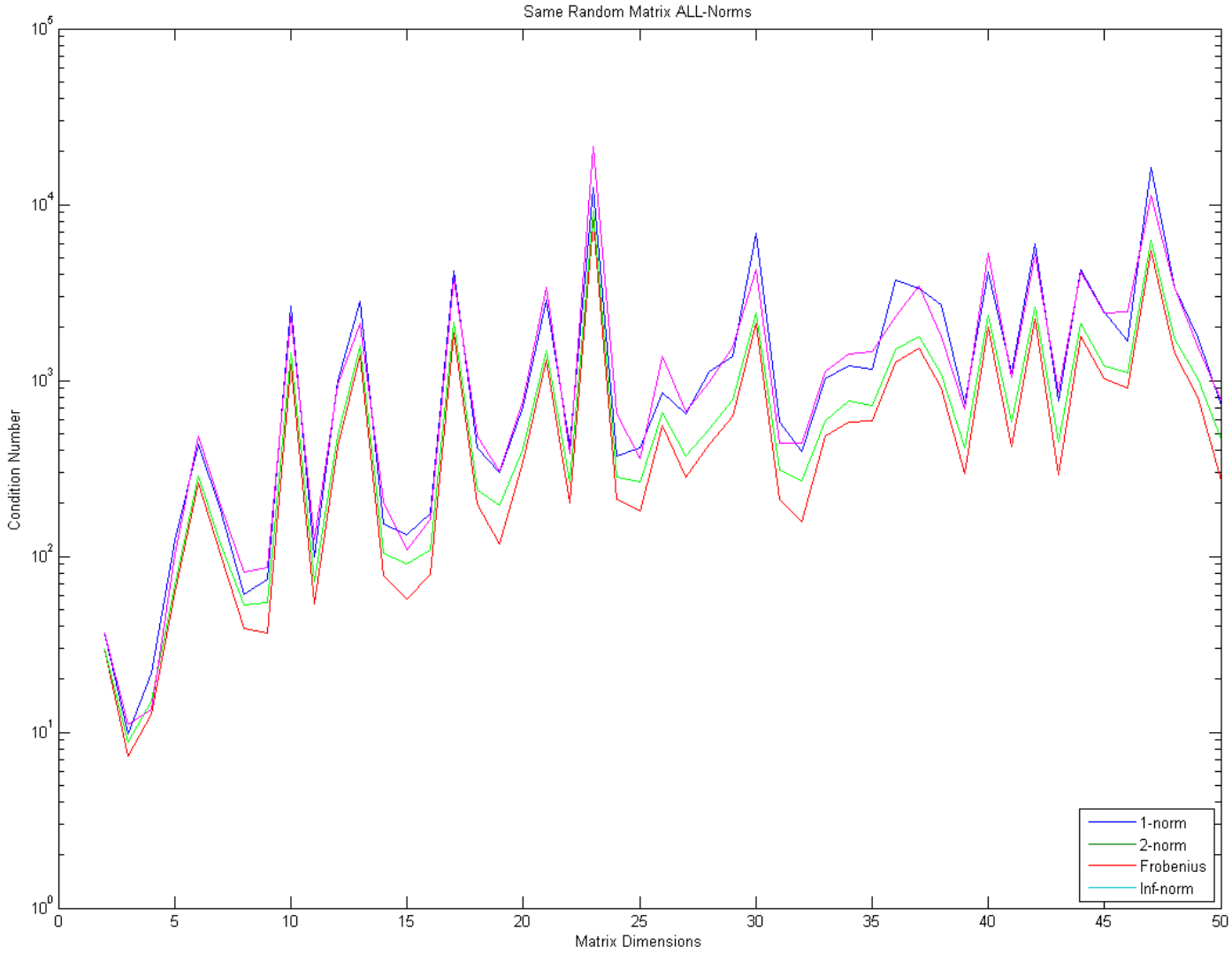



1-5 random matrices per dimension of 1-2000




The basic pattern continues for a NORMAL random distribution.
Above we have 1-1000 dimensions and 2-random matrix per set.

This is in agreement with Alan Edelman 1988 paper on SIAM (vol. 9 no. 4) where
'the expected value of the **log** of the 2-norm condition number is asymptotic to **log n** as n->∞"
Note n is the dim of the matrix, and the y scale is log y. The curve is close to a log plot.




Above we see that it seems that the Frobenius norm (red) seems to be a lower bound for all the other condition numbers. The others seem to interlace.

# Theorems and Proofs

Proposition:

For any matrix $A_n$ of dim n, and let $A_{n+1}$ be a matrix that includes $A_n$ for the $1^{st}$ nxn-dimensions, then the condition number of $A_{n+1}$ can be computed based on the cond number of $A_n$ and the vectors in the n+1 position and the last diagonal number in position n+1,n+1

# Definition of the Frobenius Condition Number

Definition 1.1 The Frobenius norm $||.||$ or $||.||_F$ for real/complex matrices $A_{n+1} = (a_{i,j})$ of dim n+1 is:

$$|| A_{n+1} ||_F = || A_{n+1} || = || (a_{i,j}) || = \sqrt{\sum_{i=1}^{n+1} \sum_{j=1}^{n+1} | a_{i,j} |^2}$$

Definition 1.2: The (Frobenius) condition number is:

$$cond(A) = || A ||_F || A^{-1} ||_F$$

Note it's defined for real/complex numbers

Here we assume the Frobenius norm (or Hilbert-Schmidt or the 2-norm).

# The Partition of a square Matrix into a 2x2 Block Matrix

We will see this procedure many times in this research. Sow we define this mapping via a formal function $N_{MP}()$.

Defintion 1.10: The 'Natural Matrix Partition' Function $N_{MP}$:

$N_{MP}: \quad M(n+1) \rightarrow MP(2)$

Where $M(n+1)$ is the set of real/complex square matrices of dim=n+1.

Where $MP(2)$ is the set of real/complex 2x2 block matrices.

The function $N_{MP}$ maps any (n+1)x(n+1) matrix into what we call a 'natural' or canonical 2x2 block matrix as follows:

Let $A_{n+1} = (b_{i,j})$ be a (n+1)x(n+1) matrix where i,j=1..n+1.

Definition 1.11: $A_n = (a_{i,j})$ as n x n matrix respectively where i,j=1..n such that:

$\forall i, j$ from 1 *to* $n$ we have: $a_{i,j} = b_{i,j}$. Sometimes we use $A_n = (b_{i,j})$ where i,j=1..n.

We use subset notation and denote this by: $A_n \subseteq A_{n+1}$ since $A_n$ is 'contained' in $A_{n+1}$.

Next we define the 'edge' vectors (to the right and below the block matrix $A_n$).

Definition 1.12: The 'edge vector#1' is: $V1_n = V_1 = (b_{i,n+1})$ where $i = 1$ to n

Definition 1.13: The 'edge vector#2' is: $V2_n = V_2 = (b_{n+1,j})$ where $j = 1$ to n

Definition 1.14 the 'last diagonal element': $D_n = (b_{n+1,n+1}) = b_{n+1,n+1}$ a 1 x 1 matrix/scalar.

Note $D_n$ is not part of any edge vector $V1_n, V2_n$ or $A_n$.

Then we define the Natural Matrix partition function $N_{MP}$ :

Defintion 1.0: $N_{MP}(A_{n+1}) = \begin{pmatrix} A_n & V1_n \\ V2_n & D_n \end{pmatrix} = (A_n, V1_n;\ V2_n, D_n)$

where semicolon means a new row.

We converted a matrix with $(n+1)^2$ elements into a 2x2 with $n^2, n, n$ and 1 element matrix. Then we can recursively continue with:

$N_{MP}(A_n) = (A_{n-1}, V1_{n-1};\ V2_{n-1}, D_{n-1})$

$N_{MP}(A_{n-1}) = (A_{n-2}, V1_{n-2};\ V2_{n-2}, D_{n-2})$

...

$N_{MP}(A_2) = (A_1, V1_1;\ V2_1, D_1)$

Now $A_2$ is 2x2 and then $A_1 = b_{1,1}$, $D_1 = b_{2,2}$, $V1_1 = b_{1,2}$, $V2_1 = b_{2,1}$.

Note that we have decreased the dim of $A_n$ by 1, with 2n+1 less elements than $A_{n+1}$. Thus: $\dim(A_{n+1}) - \dim(A_n) = (n+1)^2 - n^2 = 2n+1$,

This is due to $A_n$ not including the two edge vectors and the last diagonal.

Summary of definition 1.0 of the function $N_{MP}$:

$A_{n+1} = (b_{i,j})$ is a (n+1)x(n+1) real or complex matrix.

$A_n = (b_{i,j})$ is a nxn matrix for i,j=1..n.

Where: $A_n \subseteq A_{n+1}$

$V1_n = (b_{i,n+1})$ is a nx1 matrix (or edge column vector $V_1$)

$V2_n = (b_{n+1,j})$ is a 1xn matrix (or edge row vector $V_2$)

$D_n = (b_{n+1,n+1}) = b_{n+1,n+1}$ is a 1x1 matrix/scalar.

Then the 'Natural Matrix partition' function $N_{MP}$ is:

$$N_{MP}(A_{n+1}) = \begin{pmatrix} A_n & V1_n \\ V2_n & D_n \end{pmatrix} = \begin{pmatrix} (b_{i,j})_{i,j=1..n} & (b_{i,n+1})_{i=1..n} \\ (b_{n+1,j})_{j=1..n} & (b_{n+1,n+1}) \end{pmatrix}$$

# Formal Presentation of Definitions and Theorems

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Definition 1.1 The Frobenius norm $||.||$ or $||.||_F$ for real/complex matrices $A_{n+1} = (b_{i,j})$ of dimension n+1 is:

$$|| A_{n+1} ||_F = || A_{n+1} || = || (b_{i,j}) || = \sqrt{\sum_{i=1}^{n+1} \sum_{j=1}^{n+1} | b_{\mathrm{i,j}} |^2}$$

Definition 1.2: The (Frobenius) condition number is: $cond(A) = || A ||_F || A^{-1} ||_F$

Where $A$ is a real/complex matrix and $|| A ||_F$ is a matrix norm of $A$.

Here we assume the Frobenius norm (or Hilbert-Schmidt or 2-norm).

Let $A_n$, $A_{n+1}$ be nxn, (n+1)x(n+1) matrices respectively $(a_{i,j})$ and $(b_{i,j})$.

Let $A_n$ *and* $A_{n+1}$ be real/complex matrices such that $A_n$ is contained in $A_{n+1}$, or more precisely: $\forall i, j$ from 1 *to* $n$ we have: $a_{i,j} = b_{i,j}$, denoted $A_n \subseteq A_{n+1}$.

Separate the 1st upper nxn terms from the nx1,1xn and the diagonal $b_{n+1,n+1}$:

$$|| A_{n+1} || = \sqrt{\sum_{i=1}^{n+1} \sum_{j=1}^{n+1} | b_{\mathrm{i,j}} |^2} = \sqrt{\sum_{i=1}^{n} \sum_{j=1}^{n} | b_{\mathrm{i,j}} |^2 + \sum_{i=1}^{n} \sum_{j=n+1} | b_{\mathrm{i,j}} |^2 + \sum_{i=n+1} \sum_{j=1}^{n} | b_{\mathrm{i,j}} |^2 + | b_{n+1,n+1} |^2}$$

Next we note that all terms in the square root are positive, thus

$$\| A_{n+1} \|_F = \sqrt{\sum_{i=1}^{n}\sum_{j=1}^{n} | b_{i,j} |^2 + \sum_{i=1}^{n}\sum_{j=n+1} | b_{i,j} |^2 + \sum_{i=n+1}\sum_{j=1}^{n} | b_{i,j} |^2 + | b_{n+1,n+1} |^2} \geq \sqrt{\sum_{i=1}^{n}\sum_{j=1}^{n} | b_{i,j} |^2}$$

next since $(a_{i,j}) = A_n \subseteq A_{n+1} = (b_{i,j})$: $\forall i, j$ from 1 *to* $n$ we have: $a_{i,j} = b_{i,j}$

$$\sqrt{\sum_{i=1}^{n}\sum_{j=1}^{n} | b_{i,j} |^2} = \sqrt{\sum_{i=1}^{n}\sum_{j=1}^{n} | a_{i,j} |^2} = \| A_n \|$$ by defintion of the Frobenius norm.

then

$$\| A_{n+1} \|_F = \sqrt{\sum_{i=1}^{n}\sum_{j=1}^{n} | b_{i,j} |^2 + \sum_{i=1}^{n}\sum_{j=n+1} | b_{i,j} |^2 + \sum_{i=n+1}\sum_{j=1}^{n} | b_{i,j} |^2 + | b_{n+1,n+1} |^2} \geq \| A_n \|$$

then we have proved:

Theorem 1.0:

If $A_n \subseteq A_{n+1} \Rightarrow \| A_n \|_F \leq \| A_{n+1} \|_F$

This result is natural for Frobenius norms, but not for all other norms.

But since there is an equivalence among norms, we use $\| \circ \|_F$ for cond(A).

Theorem 2.0:

Let $M=(m_{ij})$ be a real/complex nxn matrix. A,B,C,D are block sub-matrices of M,

A, D square matrices, then the norm of M is the norm of the norms of the 2x2 block matrix:

If $M=\begin{bmatrix} A & B \\ C & D \end{bmatrix}$ then the Frobenius norm of M is: $\|M\|_F = \left\| \begin{bmatrix} \|A\|_F & \|B\|_F \\ \|C\|_F & \|D\|_F \end{bmatrix} \right\|_F$

Proof:

Denote the (four) index partitions of blocks A,B,C and D by the letters k,l,m,n,p,q,r,s in:

A-(k,l) B-(m,n) C-(p,q) and D-(r,s) where they partition (x,y) in the ranges: 1..n, 1..n.

Since the indexes form a partition of (1..n,1..n) for A,B,C and D then:

$$\|M\|_F = \sqrt{\sum_{i=1..n}\sum_{j=1..n} |m_{ij}|^2} = \sqrt{\sum_k\sum_l |m_{ij}|^2 + \sum_m\sum_n |m_{mn}|^2 + \sum_p\sum_q |m_{pq}|^2 + \sum_r\sum_s |m_{rs}|^2}$$

$$\sqrt{\|A\|_F^2 + \|B\|_F^2 + \|C\|_F^2 + \|D\|_F^2} = \left\| \begin{bmatrix} \|A\|_F & \|B\|_F \\ \|C\|_F & \|D\|_F \end{bmatrix} \right\|_F$$ or a norm of norms.

Note that the last is a 2x2 matrix whose elements are the norms of A,B,C,D.

Note that for 2x2 matrix: $\left\| \begin{matrix} a & b \\ c & d \end{matrix} \right\|_F = \sqrt{|a|^2 + |b|^2 + |c|^2 + |d|^2}$, we had $a=\|A\|_F$ etc.

Next, since $\|.\|_F = \|.\|_2$ or the p-norm for p=2, then we see Thorem 2.0 is a special case.

Definition 1.3 The '*p*-norm' for real $p \geq 1$ is $\|M\|_p = \|(m_{ij})\|_p = \left(\sum_{i=1..n} \sum_{j=1..n} |m_{ij}|^p\right)^{1/p}$

Theorem 3.0:

Let $M = (m_{ij})$ a real/complex nxn matrix in block form with A,D are square matrices :

If $M = \begin{bmatrix} A & B \\ C & D \end{bmatrix}$ then the p-norm of M is: $\|M\|_p = \left\| \begin{bmatrix} \|A\|_p & \|B\|_p \\ \|C\|_p & \|D\|_p \end{bmatrix} \right\|_p$

Proof:

Denote the (four) index *partitions* of blocks A,B,C and D by the letters:

A-(k,l) B-(m,n) C-(p,q) and D-(r,s) where they partition 1..n, 1..n.

Since the indexes form a partition of (1..n,1..n) for A,B,C and D and since p(1/p)=1:

$$\|M\|_p = \left( \sum_{i=1..n} \sum_{j=1..n} |m_{ij}|^p \right)^{1/p} = \left( \sum_k \sum_l |m_{ij}|^p + \sum_m \sum_n |m_{mn}|^p + \sum_p \sum_q |m_{pq}|^p + \sum_r \sum_s |m_{rs}|^p \right)^{1/p}$$

$$\left( \left( \left( \sum_k \sum_l |m_{ij}|^p \right)^{1/p} \right)^p + \left( \left( \sum_m \sum_n |m_{mn}|^p \right)^{1/p} \right)^p + \left( \left( \sum_p \sum_q |m_{pq}|^p \right)^{1/p} \right)^p + \left( \left( \sum_r \sum_s |m_{rs}|^p \right)^{1/p} \right)^p \right)^{1/p}$$

$$= \left( \|A\|_p^{\,p} + \|B\|_p^{\,p} + \|C\|_p^{\,p} + \|D\|_p^{\,p} \right)^{1/p} = \left\| \begin{bmatrix} \|A\|_p & \|B\|_p \\ \|C\|_p & \|D\|_p \end{bmatrix} \right\|_p$$

Note that the last is a $2x2$ matrix whose elements are the $p$-*norms* of A,B,C,D.

We need the following well-known result for the inverse of block-matrices :

Theorem 4.0:

If A,B,C,D are matrix sub-blocks of arbitrary size, A,D square matrices, then:

To simplify the formula lets define (assume nonsingular) $M$:

$$M = (D - CA^{-1}B)^{-1}$$

Then:

$$\begin{pmatrix} A & B \\ C & D \end{pmatrix}^{-1} = \begin{pmatrix} A^{-1} + A^{-1}BMCA^{-1} & -A^{-1}BM \\ -MCA^{-1} & M \end{pmatrix}$$

Proof:

Available in the literature.

Theorem 5.0:

If $\|.\|$ is the Frobenius norm and $A_{n+1}$ is a real/complex square matrix then:

$$\|A_{n+1}\| = \sqrt{\|A_n\|^2 + \|V_1\|^2 + \|V_2\|^2 + |b_{n+1,n+1}|^2} \quad \text{and}$$

$$\text{cond}(A_{n+1}) = \sqrt{\|A_n\|^2 + \|V_1\|^2 + \|V_2\|^2 + |b_{n+1,n+1}|^2} \circ \|A_{n+1}^{-1}\|$$

Proof:

Let: $A = A_n, B = V_1, C = V_2,\ D = D_n,$ where: $V_1 = V1_n = (b_{i,n+1}),\ V_2 = V2_n = (b_{n+1,j})$

where $i, j = 1$ to n. $D_n = (b_{n+1,n+1}) = b_{n+1,n+1}$ a 1x1 matrix.

Since $A_n \subseteq A_{n+1}$ we can express $A_{n+1}$ as a block matrix: $A_{n+1} = \begin{pmatrix} A_n & V_1 \\ V_2 & b_{n+1,n+1} \end{pmatrix}$

Using Theorem 2.0: $\|A_{n+1}\| = \left\| \begin{pmatrix} A_n & V_1 \\ V_2 & b_{n+1,n+1} \end{pmatrix} \right\| = \left\| \begin{pmatrix} \|A_n\| & \|V_1\| \\ \|V_2\| & \|b_{n+1,n+1}\| \end{pmatrix} \right\|$

Then: $\|A_{n+1}\| = \sqrt{\|A_n\|^2 + \|V_1\|^2 + \|V_2\|^2 + \|b_{n+1,n+1}\|^2}$

now: $\|b_{n+1,n+1}\|^2 = (\sqrt{|b_{n+1,n+1}|^2})^2 = |b_{n+1,n+1}|^2,$ by definition:

$$\text{cond}(A_{n+1}) = \|A_{n+1}\| \circ \|A_{n+1}^{-1}\| = \sqrt{\|A_n\|^2 + \|V_1\|^2 + \|V_2\|^2 + |b_{n+1,n+1}|^2} \circ \|A_{n+1}^{-1}\|$$

In theorem 5.0 we got the norm of $A_{n+1}$ based on $A_n$, now we get the norm of $A_{n+1}^{-1}$.

Theorem 6.0:

Let $M = ( b_{n+1,n+1} - V_2 A_n^{-1} V_1 )^{-1}$, and the Identity nxn matrix: $I_n = (\delta_{ij})$ then:

$$\| A_{n+1}^{-1} \|_F = \sqrt{\| A_n^{-1}(I_n + V_1 M V_2 A_n^{-1}) \|^2 + \| A_n^{-1} V_1 M \|^2 + \| M V_2 A_n^{-1} \|^2 + \| M \|^2}$$

Proof:

Assume $A_n \subseteq A_{n+1}$ and $A = A_n$, $B = V_1$, $C = V_2$, D=$b_{n+1,n+1}$ as defined in Theorem 4.0 for the inverse of a block 2x2 matrix, where we define: $M = (D - CA^{-1}B)^{-1}$, then:

$$\begin{pmatrix} A & B \\ C & D \end{pmatrix}^{-1} = \begin{pmatrix} A^{-1} + A^{-1}BMCA^{-1} & -A^{-1}BM \\ -MCA^{-1} & M \end{pmatrix}$$ ,where: $M = ( b_{n+1,n+1} - V_2 A_n^{-1} V_1 )^{-1}$,

we know by Theorem 4.0 the inverse of $A_{n+1}^{-1}$ expressed as a block matrix is:

$$A_{n+1}^{-1} = \begin{pmatrix} A_n & V_1 \\ V_2 & b_{n+1,n+1} \end{pmatrix}^{-1} = \begin{pmatrix} A_n^{-1} + A_n^{-1} V_1 M V_2 A_n^{-1} & -A_n^{-1} V_1 M \\ -M V_2 A_n^{-1} & M \end{pmatrix}$$

With the application of Theorem 3.0 we obtain:

$$\| A_{n+1}^{-1} \|_F = \left\| \begin{pmatrix} A_n & V_1 \\ V_2 & b_{n+1,n+1} \end{pmatrix}^{-1} \right\| = \text{ Next we replace the inverse by the block inverse:}$$

$$\left\| \begin{pmatrix} A_n^{-1} + A_n^{-1} V_1 M V_2 A_n^{-1} & -A_n^{-1} V_1 M \\ -M V_2 A_n^{-1} & M \end{pmatrix} \right\| = \left\| \begin{pmatrix} \| A_n^{-1} + A_n^{-1} V_1 M V_2 A_n^{-1} \| & \| -A_n^{-1} V_1 M \| \\ \| -M V_2 A_n^{-1} \| & \| M \| \end{pmatrix} \right\|$$

Where: $M = (D - CA^{-1}B)^{-1} = (b_{n+1,n+1} - V_2 A_n^{-1} V_1)^{-1}$ , which is a 1x1 scalar/matrix.

Now $\|\alpha A\| = |\alpha| * \| A\|$, thus -1 signs are eliminated next and we factor $A_n^{-1}$.

$$\| A_{n+1}^{-1} \| = \left\| \begin{pmatrix} \| A_n^{-1} + A_n^{-1} V_1 M V_2 A_n^{-1} \| & \| A_n^{-1} V_1 M \| \\ \| M V_2 A_n^{-1} \| & \| M \| \end{pmatrix} \right\| =$$

$$\| A_{n+1}^{-1} \|_F = \sqrt{\| A_n^{-1} (I_n + V_1 M V_2 A_n^{-1}) \|^2 + \| A_n^{-1} V_1 M \|^2 + \| M V_2 A_n^{-1} \|^2 + \| M \|^2}$$

*QED*

Definition 1.15: $b_0 = b_{n+1,n+1}$ the last diagonal of $A_{n+1}$, a scalar.

Definition 1.16: $k_0 = V_2 A_n^{-1} V_1$ a matrix product of 1xn, nxn, nx1=1x1 a scalar.

Definition 1.16: $\mathrm{M} = 1/(b_0 - k_0) = (b_{n+1,n+1} - V_2 A_n^{-1} V_1)^{-1} = m_0$ a scalar.

Definition 1.17: $V_1 V_2 = v_0$ a scalar, since $V_1$ and $V_2$ are 1xn, nx1, $V_1 V_2$ is 1x1.

Definition 1.17: $L_n = I_n + m_0 v_0 A_n^{-1}$ where $I_n$ is the nxn Identity matrix.

Theorem 6.1:

$$\| A_{n+1}^{-1} \|_F = \sqrt{\| A_n^{-1} L_n \|^2 + | m_0 |^2 (\| A_n^{-1} V_1 \|^2 + \| V_2 A_n^{-1} \|^2 + 1)}$$

Proof:

Since M is a scalar named $m_0$, and $V_1 V_2 = v_0$ is also a scalar, then

$V_1 M V_2 = V_1 m_0 V_2 = m_0 V_1 V_2 = m_0 v_0$. We use this in: $A_n^{-1}(I + (V_1 M V_2) A_n^{-1})$ below.

Note: $(\sqrt{m_0^{\,2}})^2 = |m_0|^2$, then:

$$\| A_{n+1}^{-1} \|_F = \sqrt{\| A_n^{-1}(I_n + V_1 M V_2 A_n^{-1}) \|^2 + \| A_n^{-1} V_1 M \|^2 + \| M V_2 A_n^{-1} \|^2 + \| M \|^2}$$

$$\| A_{n+1}^{-1} \|_F = \sqrt{\| A_n^{-1}(I_n + m_0 v_0 A_n^{-1}) \|^2 + | m_0 |^2 (\| A_n^{-1} V_1 \|^2 + \| V_2 A_n^{-1} \|^2 + 1)}$$

$$| A_{n+1}^{-1} \|_F = \sqrt{\| A_n^{-1} L_n \|^2 + | m_0 |^2 (\| A_n^{-1} V_1 \|^2 + \| V_2 A_n^{-1} \|^2 + 1)}$$

The new central formula 6.2 below computes directly the cond($A_{n+1}$) based on the submatrices of $A_{n+1}$ : $A_n, V_1, V_2, b_0$ and computed matrices based on $A_n^{-1}, V_1, V_2, b_0, I_n$, while the definition of condition number uses the norms of $A_{n+1}$ and the norm of $A_{n+1}^{-1}$. We have an exact expression for the condition number of a matrix $A_{n+1}$ based only on it's submatrix $A_n$, the two edge vectors $V_1, V_2$ and the last diagonal: $b_0 = b_{n+1,n+1}$.
The other matrices or scalars are derived from these:
$A_n^{-1}$ from $A_n$; $k_0 = V_2 A_n^{-1} V_1$ from $V_2, A_n^{-1}, V$; $m_0$ from $b_0, k_0$; $L_n$ from $I_n, m_0, v_0, A_n^{-1}$.

Theorem 6.2

Let: $b_0 = b_{n+1,n+1}$; $k_0 = V_2 A_n^{-1} V_1$; $m_0 = 1/(b_0 - k_0)$, $V_1 V_2 = V$, $L_n = I_n + m_0 V A_n^{-1}$

Then: $\text{cond}(A_{n+1}) = \| A_{n+1} \| \circ \| A_{n+1}^{-1} \| =$

$$\sqrt{\| A_n \|^2 + \| V_1 \|^2 + \| V_2 \|^2 + | b_0 |^2} \sqrt{\| A_n^{-1} L_n \|^2 + | m_0 |^2 (\| A_n^{-1} V_1 \|^2 + \| V_2 A_n^{-1} \|^2 + 1)}$$

Proof:

Follows directly from the definition of condition number and Theorems 5.0, 6.1.

We have simply multiplied the results for $\| A_{n+1} \|$ and for $\| A_{n+1}^{-1} \|$ based on $A_n$.

How each part of the submatrix influence the cond number?

To convert to a sum the value of cond($A_{n+1}$) we square the factors $\| A_{n+1} \|^2 \| A_{n+1}^{-1} \|^2$ :

Let $b_0 = b_{n+1,n+1}, V = V_1 V_2, k_0 = V_2 A_n^{-1} V_1, m_1 = | m_0 | = | 1/(b_0 - k_0) |,\ L_n = I_n + m_0 V A_n^{-1}$

Theorem 7.0:

$$\text{cond}(A_{n+1})^2 = \| A_{n+1} \|^2 \| A_{n+1}^{-1} \|^2 =$$

$$(\| A_n \|^2 + \| V_1 \|^2 + \| V_2 \|^2 + | b_0 |^2)(\| A_n^{-1} L_n \|^2 + m_1^2 \| A_n^{-1} V_1 \|^2 + m_1^2 \| V_2 A_n^{-1} \|^2 + m_1^2)$$

$=$

$$\| A_n \|^2 \| A_n^{-1} L_n \|^2 + m_1^2 \| A_n \|^2 \| A_n^{-1} V_1 \|^2 + m_1^2 \| A_n \|^2 \| V_2 A_n^{-1} \|^2 + m_1^2 \| A_n \|^2 +$$

$$\| V_1 \|^2 \| A_n^{-1} L_n \|^2 + m_1^2 \| V_1 \|^2 \| A_n^{-1} V_1 \|^2 + m_1^2 \| V_1 \|^2 \| V_2 A_n^{-1} \|^2 + m_1^2 \| V_1 \|^2 +$$

$$\| V_2 \|^2 \| A_n^{-1} L_n \|^2 + m_1^2 \| V_2 \|^2 \| A_n^{-1} V_1 \|^2 + m_1^2 \| V_2 \|^2 \| V_2 A_n^{-1} \|^2 + m_1^2 \| V_2 \|^2 +$$

$$b_0^2 \| A_n^{-1} L_n \|^2 + m_1^2 b_0^2 \| A_n^{-1} V_1 \|^2 + m_1^2 b_0^2 \| V_2 A_n^{-1} \|^2 + m_1^2 b_0^2$$

cond($A_{n+1}$) $= f(A_n, V_1, V_2, b_0)$ can be considered as depending on the $\|.\|_F$ norm,

on the submatrix $A_n$; $A_n^{-1}$ depends on $A_n$; $V$ depends on $V_1, V_2$; $k_0$ depends on $V_1, V_2, A_n$;

$m_0$ depends on $b_{n+1,n+1}, V_1, V_2$ and $A_n$; $E_n$ depends on $m_0, V, A_n$;

Then: $f(A_n, V_1, V_2, b_0) == \sqrt{\| A_{n+1} \|^2 \| A_{n+1}^{-1} \|^2} = \sqrt{\text{cond}(A_{n+1})^2} = \text{cond}(A_{n+1})$.

Next We plot the value of the condition numbers from the calculated and from the new formula based on submatrices.
This ‘verifies’ the correctnes of the formula:

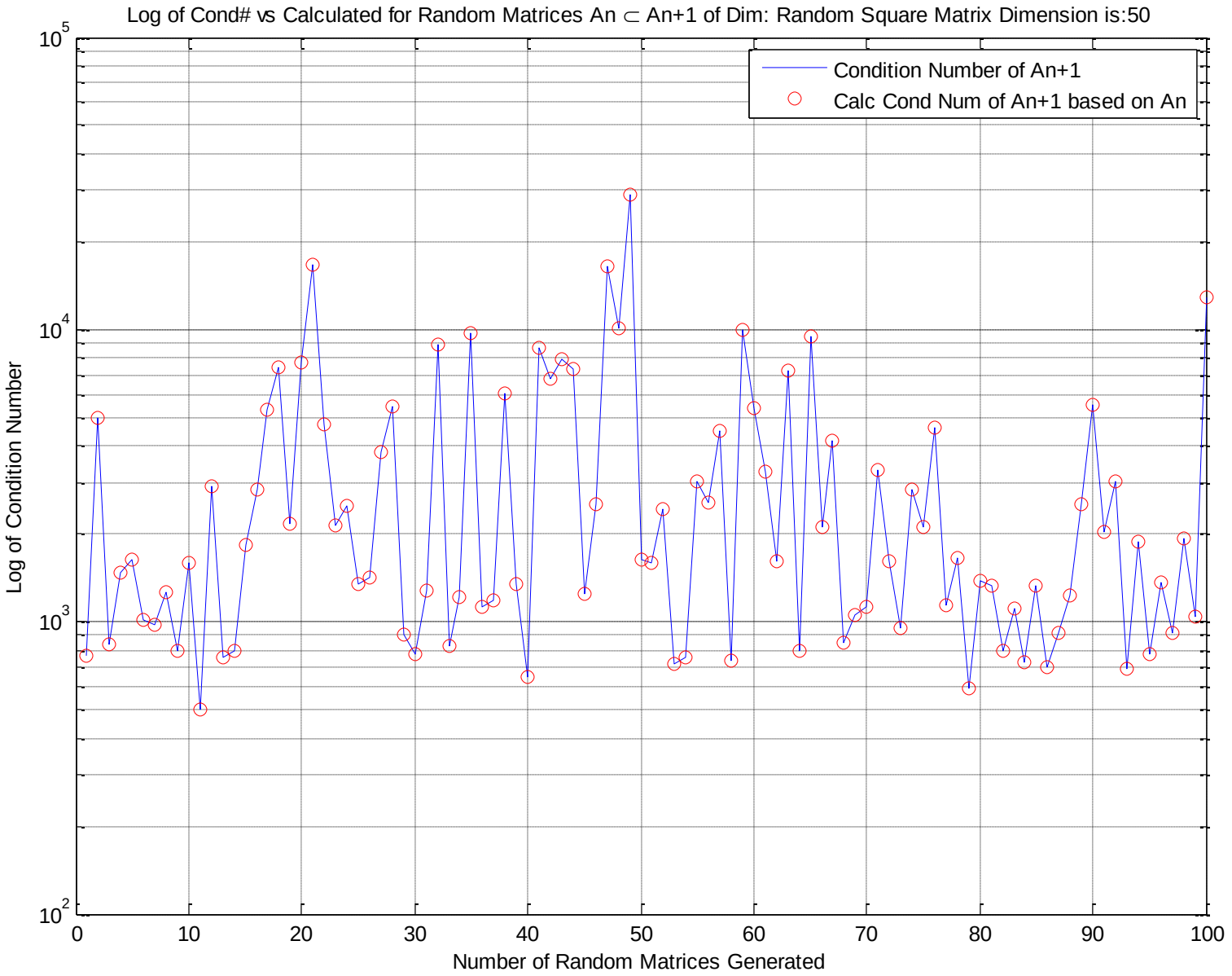
Theorem 6.2

Let: $b_0 = b_{n+1,n+1}$; $k_0 = V_2 A_n^{-1} V_1$; $m_0 = 1/(b_0 - k_0)$, $V_1 V_2 = V$, $L_n = I_n + m_0 V A_n^{-1}$

Then: $\mathrm{cond}(A_{n+1}) = \| A_{n+1} \| \circ \| A_{n+1}^{-1} \| =$

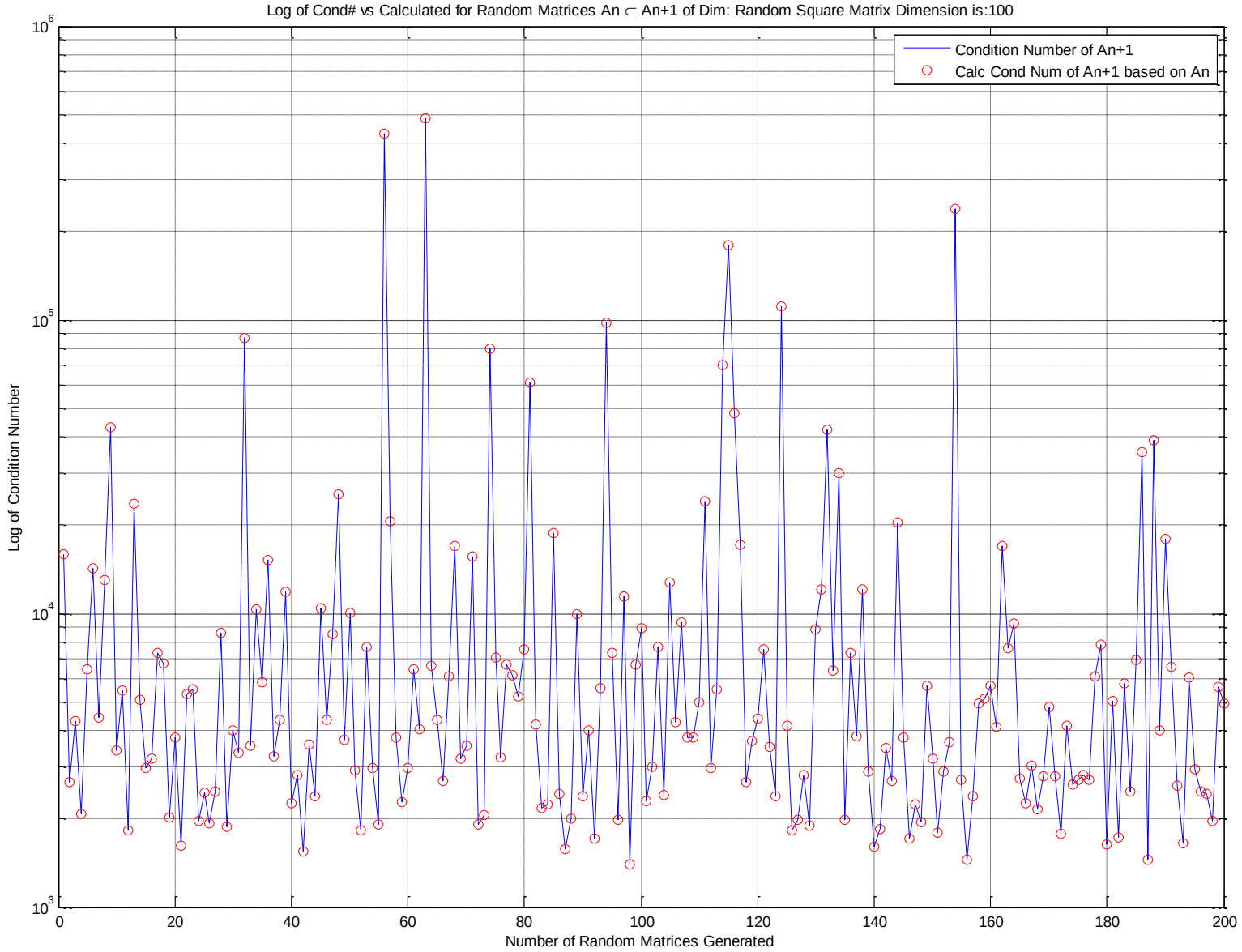
$$\sqrt{\| A_n \|^2 + \| V_1 \|^2 + \| V_2 \|^2 + | b_0 |^2} \sqrt{\| A_n^{-1} L_n \|^2 + | m_0 |^2 (\| A_n^{-1} V_1 \|^2 + \| V_2 A_n^{-1} \|^2 + 1)}$$

# 100 Trials of 50x50 Matrices




The plot per each trial random matrix the cond number vs calculated values

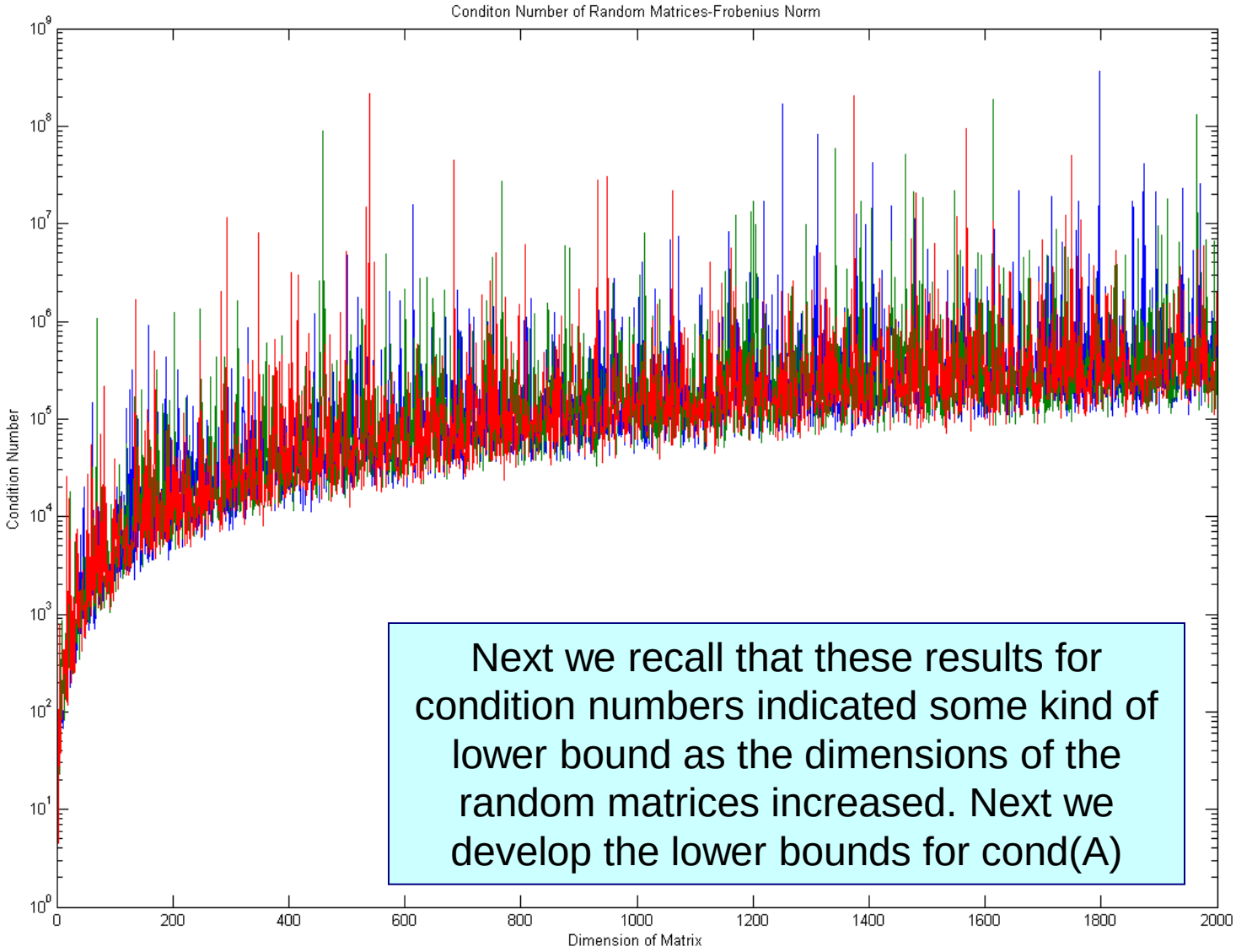
# 200 Trials of 100x100 Matrices




The plot per each trial random matrix the cond number vs calculated values

# Condition Number several lower bounds

Since we have an equation for the cond($A_{n+1}$) and for the related norms, we can use these to produce three inequalities for the norm and three for the condition number, among many other possibilities




1-5 random matrices per dimension of 1-2000

# Numerous Inequalities Resulting From the Main Equation in Theorem 6.1

$$\| A_{n+1}^{-1} \| = \sqrt{\| A_n^{-1}(I_n + m_0 V A_n^{-1}) \|^2 + | m_0 |^2 \| A_n^{-1} V_1 \|^2 + | m_0 |^2 \| V_2 A_n^{-1} \|^2 + | m_0 |^2}$$

Define: $L_n = I_n + m_0 V A_n^{-1}$ and $\text{cond}(A_{n+1}) = \| A_{n+1} \| \circ \| A_{n+1}^{-1} \|$

And common norms properties:

$\|AB\| \le \|A\| \circ \|B\|$, $\|A+B\| \le \|A\| + \|B\|$, $\|\alpha A\| = |\alpha| \circ \| A \|$, $\| A \| \ge 0$

By previous results where : $A_n \subseteq A_{n+1}$ we obtained the equality:

$$\| A_{n+1}^{-1} \| = \sqrt{\| A_n^{-1}(I_n + m_0 V A_n^{-1}) \|^2 + | m_0 |^2 \| A_n^{-1} V_1 \|^2 + | m_0 |^2 \| V_2 A_n^{-1} \|^2 + | m_0 |^2}$$

Since all terms inside are positive, we can eliminate one or more terms.

Note below the importance of the value: $m_0 = (b_{n+1,n+1} - V_2 A_n^{-1} V_1)^{-1}$ and using **:

IE1.10 $\| A_{n+1}^{-1} \| \geq | m_0 |$

IE1.12 $\| A_{n+1}^{-1} \| \geq \sqrt{\| A_n^{-1}(I_n + m_0 V A_n^{-1}) \|^2} = \| A_n^{-1}(I + m_0 V A_n^{-1}) \|$

IE1.14 $\| A_{n+1}^{-1} \| \geq \| A_n^{-1} \| \circ \| I_n + m_0 V A_n^{-1} \|$

IE1.16 $\| A_{n+1}^{-1} \| \geq \| A_n^{-1} \| + | m_0 | \circ \| V A_n^{-1} \|$

IE1.20 $\| A_{n+1}^{-1} \| \geq \sqrt{\| A_n^{-1}(I_n + m_0 V A_n^{-1}) \|^2 + m_0^2 \| A_n^{-1} V_1 \|^2 + m_0^2 \| V_2 A_n^{-1} \|^2}$

IE1.30 $\| A_{n+1}^{-1} \| \geq | m_0 | \sqrt{\| A_n^{-1} V_1 \|^2 + \| V_2 A_n^{-1} \|^2 + 1}$

IE1.41 $\| A_{n+1}^{-1} \| \geq | m_0 | \circ \| A_n^{-1} V_1 \|$

IE1.42 $\| A_{n+1}^{-1} \| \geq | m_0 | \circ \| V_2 A_n^{-1} \|$

Each of these inequalities may be useful depending on the values of $A_n^{-1}, V_1, V_2, m_0$.

** Used: $\|AB\| \leq \|A\| \circ \|B\|$, $\|A+B\| \leq \|A\| + \|B\|$, $\|\alpha A\| = |\alpha| \circ \| A \|$, $\| A \| \geq 0$

Recall the useful nxn matrix used before: $L_n = I_n + m_0 V A_n^{-1}$,

next we multiply by $\| A_{n+1} \| \geq 0$ and use: $\text{cond}(A_{n+1}) = \| A_{n+1} \| \circ \| A_{n+1}^{-1} \|$

and since all terms inside are positive, we can eliminate one or more to obtain:

IE1.10c $\text{cond}(A_{n+1}) \geq \| A_{n+1} \| \sqrt{\| A_n^{-1}(I_n + m_0 V A_n^{-1}) \|^2} = \| A_{n+1} \| \circ \| A_n^{-1} L_n \|$

IE1.20c $\text{cond}(A_{n+1}) \geq \| m_0 \| \circ \| A_{n+1} \|$

IE1.30c $\text{cond}(A_{n+1}) \geq \| A_{n+1} \| \sqrt{\| A_n^{-1} L_n \|^2 + m_0^2 (\| A_n^{-1} V_1 \|^2 + \| V_2 A_n^{-1} \|^2)}$

IE1.40c $\text{cond}(A_{n+1}) \geq \| m_0 \| \circ \| A_{n+1} \| \sqrt{\| A_n^{-1} V_1 \|^2 + \| V_2 A_n^{-1} \|^2 + 1}$

IE1.41c $\text{cond}(A_{n+1}) \geq \| m_0 \| \circ \| A_{n+1} \| \circ \| A_n^{-1} V_1 \|$

IE1.42c $\text{cond}(A_{n+1}) \geq \| m_0 \| \circ \| A_{n+1} \| \circ \| V_2 A_n^{-1} \|$

Some inequalities are skipped since they included $\| A_{n+1} \| \circ \| A_{n+1}^{-1} \|$ on the right side.

Each of these inequalities may be useful, depending on $A_n^{-1}, A_n, V_1, V_2, V, L_n, m_0, b_0$.

Thus we obtain the author's original purpose - inequalities for $\text{cond}(A_{n+1})$ based on $A_n$.

** Used: $\|AB\| \leq \|A\| \circ \|B\|$, $\|A+B\| \leq \|A\| + \|B\|$, $\|\alpha A\| = |\alpha| \circ \| A \|$, $\| A \| \geq 0$

Numerical experiments with matrices with large condition numbers seem to point to the large values being reflected in the norm of :

$$E_n = A_n^{-1}(I_n + m_0 V A_n^{-1}) = A_n^{-1} L_n$$

By previous results where : $A_n \subseteq A_{n+1}$ we obtained the equality:

$$\| A_{n+1}^{-1} \| = \sqrt{\| E_n \|^2 + | m_0 |^2 \| A_n^{-1} V_1 \|^2 + | m_0 |^2 \| V_2 A_n^{-1} \|^2 + | m_0 |^2}$$

where $b_0 = b_{n+1,n+1}, k_0 = V_2 A_n^{-1} V_1, m_0 = 1/(b_0 - k_0), V = V_1 V_2$

Therefore we try to understand better the behaviour of $\|E_n\|^2$ :

$$\| E_n \|^2 = \|A_n^{-1} L_n\|^2 \le \|A_n^{-1}\|^2 \circ \| L_n \|^2$$

$$\| L_n \|^2 = \| (I_n + m_0 V A_n^{-1}) \|^2 \le (\| I_n \| + \| m_0 V A_n^{-1} \|)^2$$

Lemma 1.0:

$$\| I_n \| = \| I_n \|_F = \sqrt{\sum_{i,j=1..n} \delta^2_{ij}} = \sqrt{\sum_{i=1..n} \delta^2_{ii}} = \sqrt{\sum_{i=1..n} 1^2} = \sqrt{n} \text{ and } \|I_n\|^2 = n$$

And note since $m_0$ is a scalar: $\| m_0 V A_n^{-1} \| = | m_0 | \ \| V A_n^{-1} \|$

$$(\| I_n \| + \| m_0 V A_n^{-1} \|)^2 = (\sqrt{n} + | m_0 | \circ \| V A_n^{-1} \|)^2 = n + 2\sqrt{n} | m_0 | \circ \| V A_n^{-1} \| + \| V A_n^{-1} \|^2$$

$$\|A_n^{-1}(I_n + m_0 V A_n^{-1})\|^2 \le \|A_n^{-1}\|^2 \circ \| (I_n + m_0 V A_n^{-1}) \|^2$$

** Used: $\|AB\| \le \|A\| \circ \|B\|$, $\|A+B\| \le \|A\| + \|B\|$, $\|\alpha A\| = |\alpha| \circ \| A \|$, $\| A \| \ge 0$

Now we follow the previous result:

$$\|A_n^{-1}(I_n + m_0 V A_n^{-1})\|^2 \le \|A_n^{-1}\|^2 \, (n + 2\sqrt{n}\,|m_0| \circ \|VA_n^{-1}\| + \|VA_n^{-1}\|^2)$$

The significance here is that we have related the dimension of the matrix $A_n = n$ to an upper bound for: $\|A_n^{-1}(I_n + m_0 V A_n^{-1})\|^2 = \|A_n^{-1} L_n\|$

IE9.1 $\|A_n^{-1}(I_n + m_0 V A_n^{-1})\|^2 \le \|A_n^{-1}\|^2 \, (n + 2\sqrt{n}\,|m_0| \circ \|VA_n^{-1}\| + \|VA_n^{-1}\|^2)$

Recall that by definition of condition number for $A_n$ : $\text{cond}(A_n) = \|A_n\| \circ \|A_n^{-1}\|$

|thus multiply IE9.1 by the positive number: $\|A_n\|^2$ on both sides:

$$\|A_n\|^2 \circ \|A_n^{-1}(I_n + m_0 V A_n^{-1})\|^2 \le \|A_n\|^2 \circ \|A_n^{-1}\|^2 \, (n + 2\sqrt{n}\,|m_0| \circ \|VA_n^{-1}\| + \|VA_n^{-1}\|^2)$$

This proves the upper bound:

Theorem 6UB:

$$\|A_n\|^2 \circ \|A_n^{-1}(I_n + m_0 V A_n^{-1})\|^2 \le \text{cond}(A_n) \circ (n + 2\sqrt{n}\,|m_0| \circ \|VA_n^{-1}\| + \|VA_n^{-1}\|^2)$$

# A Basic Point Relative to Random Matrices

- Many of us know that matrices are essential and widely used in many fields where quantitative results are needed
- Examples are Science, Engineering, Statistics, Social sciences etc
- In particular fields like Engineering we have Finite element, Dynamics and Mechanics, Fluid Dynamics were these are used in many forms (graph connectivity matrix, stresses, Navier-Stokes etc)
- Then note that ANY square matrix with real positive elements can be seen as a random matrix (all elements inside [0,1])
- That is if we multiply $A=(a_{ij})$ by it's $m=\max(a_{ij})$ then $m^{-1}A$ has all elements $0 <= a_{ij}/m <= 1$ and
- $Cond(m^{-1}A)=|m^{-1}|cond(A)$ thus we can directly compute the condition number of $m^{-1}A$ from the condition number of A,
- Thus when we study **random matrices** we are also covering the cases of many actual applications
- Thus the behavior of condition number of random matrices can be of help understanding the condition number of matrices found in the applied fields

# Observations for Special Matrices

- Each of these matrices allow us to simplify the formula for cond($A_{n+1}$)
- Diagonal (trivial)
- Tri-diagonal
- Upper or Lower triangular
- Positive definite
- LU decomposition
- QR decomposition
- Sparse
- Specific Matrices classes
- This is a work in progress (wip)

# Observations for Special Matrices
# WIP

- **Diagonal** – always the V1n=V2n=0 (edge vectors). This results in v0=V1n*V2n=0, F1n=F2n=0, k0=0 etc. All $A_k$ are also diagonal matrices for k=1..n. Trivial case.
- **Tri-diagonal** – it's similar to the diagonal but the V1n and V2n has 1 element (the last in the vector) and thus the norm is the abs value of it.
- **Upper or Lower triangular** – one of the V1n or V2n is zero.
- **Positive definite** – all the submatrices $A_k$ have $|A_k|>0$.
- **Sparse** – since it has dim=n at least one element in the edge vectors or the last diagonal element is non zero

# ‘cond’ Command in Matlab - Condition number

**cond**
Condition number with respect to inversion

**Syntax**
c = cond(X)
c = cond(X,p)

We use ‘cond(X,’fro’) or cond(X,p) were we pass p=‘fro’)

**Description**
**The *condition number* of a matrix measures the sensitivity of the solution of a system of linear equations to errors in the data**. It gives an indication of the accuracy of the results from matrix inversion and the linear equation solution.
Values of cond(X) and cond(X,p) near 1 indicate a well-conditioned matrix.
c = cond(X) returns the 2-norm condition number, the ratio of the largest singular value of X to the smallest.
c = cond(X,p) returns the matrix condition number in p-norm:
norm(X,p) * norm(inv(X),p)

**If p is...Then cond(X,p) returns the...**
1-norm condition number
2-norm condition number
'fro‘ - Frobenius norm condition number
Inf - Infinity norm condition number

**Algorithm**
The algorithm for cond (when p = 2) uses the singular value decomposition, svd.
When the input matrix is sparse, cond ignores any specified p value and calls condest.

# ‘cond’ Command in Matlab
# Condition number

| If p is... | Then cond(X,p) returns the... |
|---|---|
| 1 | 1-norm condition number |
| 2 | 2-norm condition number |
| 'fro' | Frobenius norm condition number |
| inf | Infinity norm condition number |


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